

Acute and chronic watercress supplementation attenuates exercise-induced peripheral mononuclear cell DNA damage and lipid peroxidation.

Pharmacological antioxidant vitamins have previously been investigated for a prophylactic effect against exercise-induced oxidative stress. However, large doses are often required and may lead to a state of pro-oxidation and oxidative damage. Watercress contains an array of nutritional compounds such as β -carotene and α -tocopherol which may increase protection against exercise-induced oxidative stress. The present randomised controlled investigation was designed to test the hypothesis that acute (consumption 2 h before exercise) and chronic (8 weeks consumption) watercress supplementation can attenuate exercise-induced oxidative stress. A total of ten apparently healthy male subjects (age 23 (SD 4) years, stature 179 (SD 10) cm and body mass 74 (SD 15) kg) were recruited to complete the 8-week chronic watercress intervention period (and then 8 weeks of control, with no ingestion) of the experiment before crossing over in order to complete the single-dose acute phase (with control, no ingestion). Blood samples were taken at baseline (pre-supplementation), at rest (pre-exercise) and following exercise. Each subject completed an incremental exercise test to volitional exhaustion following chronic and acute watercress supplementation or control. The main findings show an exercise-induced increase in DNA damage and lipid peroxidation over both acute and chronic control supplementation phases ($P < 0.05$ v. supplementation), while acute and chronic watercress attenuated DNA damage and lipid peroxidation and decreased H_2O_2 accumulation following exhaustive exercise ($P < 0.05$ v. control). A marked increase in the main lipid-soluble antioxidants (α -tocopherol, γ -tocopherol and xanthophyll) was observed following watercress supplementation ($P < 0.05$ v. control) in both experimental phases. These findings suggest that short- and long-term watercress ingestion has potential antioxidant effects against exercise-induced DNA damage and lipid peroxidation.